
Lossfunk Autoresearch Submission

Hormetic IB Scheduling for Persistent Object Identity

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github.com/Incharajayaram/hormetic-ib-slot

Tool: Autovoila (Lossfunk) + Claude Code (claude-sonnet-4-6)

1. Starting Research Question

The Claim

Slot-based object-centric representations trained under a **hormetic** Information-Bottleneck schedule (beta increasing progressively during training) retain object identity through occlusion more robustly than identical architectures trained at fixed beta or non-monotone beta schedules.

The Intervention is Minimal

Same architecture, same parameter count, same final beta

Only the beta(t) trajectory differs across 6 conditions

Primary metric: identity-retention accuracy under occlusion on CLEVRER and ADEPT

One-Sentence Version

A slot-based object representation trained under a hormetic IB schedule maintains object identity through occlusion more robustly than fixed-beta or non-monotone-beta baselines, measurable on CLEVRER and ADEPT against published slot-based persistence systems.

2. Autoresearch Flow

Tool

Autovoila (Lossfunk prompt) on Claude Code (claude-sonnet-4-6), YOLO mode via cco.

Session Flow

Provided the Research Question Sharpener as starting prompt

Claude scaffolded the full project in one session: Slot Attention encoder, VIB head, 6 schedule classes, training/eval pipeline, data loaders

Ran a synthetic pilot (toy Gaussian data, CPU, 100 steps) to verify the pipeline

No real CLEVRER/ADEPT runs — require A100 + ~450 GPU-hours

Customization

None to the Autovoila prompt

Research proposal was detailed enough that Claude needed very little mid-session direction

Outputs reviewed but not edited

Human Intervention Level: Low

Provided initial framing (Research Question Sharpener), reviewed output, did not touch artifact code or results.

3. Summary of Results

All numbers are from synthetic experiments (toy Gaussian data, CPU). Not from CLEVRER or ADEPT.

Condition	Identity Retention @k=2	Identity Retention @k=4	Slot Collapse
Hormetic-Cosine	0.553	0.0	75%
Hormetic-Sigmoid	0.535	0.0	75%
Random Permutation	0.368	0.0	76%
Reverse	0.348	0.0	75%
Fixed-beta (baseline)	0.287	0.0	76%
Linear	0.250	0.0	75%

AI's reading: Hormetic conditions outperform baselines at k=2. Consistent with the hypothesis.

What this actually means: see Critique slides.

4. What I Learned About the Research Question

What the AI Confirmed

The experimental design is correct and the controls are well-chosen

The code runs; the question can be operationalized

The scaffold is a genuine starting point for real experiments

What Surprised Me

The question is easier to produce *results that look like evidence* without being evidence than I expected. The numerical ordering on synthetic data matches the hypothesis directionally, but that synthetic data was never validated as a proxy for CLEVRER/ADEPT dynamics. A direction match on toy data is not a result.

The Deeper Gap

The research question is well-posed at the proposal level, but the meaningful empirical content lives entirely in the actual training runs on real data with real occlusion events. That is precisely what the auto-research system could not do. The gap between 'scaffold is correct' and 'experiment ran' is the entire scientific content of this project.

5. Critique: What the AI Got Right

The codebase is structurally clean and correct.

Component	Why it's correct
VIB loss	$L = \text{recon_loss} + \text{beta} \times \text{KL}(q(z x) p(z))$ with closed-form diagonal Gaussian KL
Hungarian matching	Correctly used for slot-to-object assignment in identity-retention evaluation
6-condition ablation	Random permutation + reverse baselines are the right controls to isolate schedule geometry
Identity-retention metric	Same slot index maps to same object pre/post occlusion at k in $\{4,8,16,32\}$ frames
Data loaders	CLEVRER + ADEPT interfaces structurally correct; would work with real datasets present
README	Honestly documents the claim, expected compute (~450 GPU-hours), and data requirements

6. Critique: What the AI Got Wrong

1. The experiments are not the experiments.

Synthetic Gaussian data is not CLEVRER or ADEPT. The claim is about visual occlusion in video. The AI ran what it could and reported it as informative. It is not.

2. 75% slot collapse invalidates all numbers.

When 3 of 4 slots degenerate, identity-retention scores measure noise routing, not object representations. The experiment failed before producing a result.

3. Identity retention at $k=4 = 0.0$ for every condition.

No condition preserved identity across 4 frames. This is a training failure. The AI did not flag it as anomalous.

4. The directional match at $k=2$ is probably noise.

Hormetic-Cosine 0.553 vs Fixed-beta 0.287 is one run on synthetic data with 75% collapse. This is exactly the kind of number that looks like a result and is nothing.

5. The paper was never written.

The caisc_2026.tex in the repo is the blank submission template, unchanged. No abstract, no methods, no results in paper form.

6. No IB-plane trajectories, no OOD evaluation.

Both listed as secondary measurements in the design. Neither was produced.

7. Reflection: Limits of Auto-Research on This Question

What the AI did well

Translated a well-specified proposal into working code fast; a human needs 2-4 weeks for the same scaffold

Picked the right components: VIB, Hungarian matching, SAVi-style temporal slot attention, spatial broadcast decoder

Designed the correct ablation structure including matched-beta controls

What kind of question would AI do better on?

A question where compute is small, data is accessible, evaluation is unambiguous, and prior literature has a clear quantitative baseline to beat. Something like: 'Does this architectural change improve ARI on CLEVRER in under 10 GPU-hours?' That is nearly fully automatable. This question requires 450 GPU-hours, two datasets, and evaluation that depends on ruling out multiple confounds.

What the AI failed at

Cannot run experiments requiring GPUs and datasets it does not have. The entire empirical content of the claim lives in 450 GPU-hours on CLEVRER/ADEPT.

Did not recognize when its own results were invalid. A 75% collapse rate and all-zero retention at $k=4$ are failure indicators, not results.

'The pipeline ran' is not 'the experiment worked.' Running without error on toy data is not evidence.

Cannot construct the interpretive argument. The novel claim requires reading IB theory, Slot Attention, and Saxe et al.'s critique together with nuance the AI cannot provide.

8. Revised Research Plan

What Stays the Same

Core claim is still worth testing; experimental design is correct
AI-generated codebase is a usable starting point

Priority 1: Fix Slot Collapse First

The 75% collapse rate must be diagnosed before any schedule comparison. Check gradient flow through the VIB head, verify KL weighting is not too aggressive in early training, consider a slot-diversity auxiliary loss. No schedule comparison is meaningful until collapse is resolved.

Priority 2: Validate the Synthetic Proxy

If fast synthetic experiments are to screen schedules, first verify that synthetic-data trends correlate with CLEVRER trends. The AI assumed this correlation; it needs to be confirmed.

Priority 3: Reduce Scope for Budget Constraints

Instead of 6 conditions x 5 seeds x 100 epochs on two datasets, a realistic version: 3 conditions (Hormetic-Sigmoid vs. Linear vs. Fixed-beta) x 3 seeds x 30 epochs on CLEVRER only, primary metric only. Roughly 10x cheaper, still tests the core claim.

Priority 4: Write the Paper

The AI can draft the paper once real results exist. This step was skipped entirely. It should be the final step, not an afterthought.

What Changed from Original Proposal

The proposal was technically sound. What changed is the ordering: collapse diagnosis must come before schedule comparison, and scope must be reduced to match available compute.

9. Summary

Item	Status
Research question	Well-posed, falsifiable, still worth testing
Codebase / scaffold	AI-generated, structurally correct, usable
Experiments (CLEVRER/ADEPT)	Not run — require GPU and real datasets
Synthetic pilot	Run but invalid: 75% collapse, 0.0 retention at k=4
Paper draft	Not written — blank template only
GitHub repo	github.com/Incharajayaram/hormetic-ib-slot

The honest one-sentence version:

The AI built the lab but could not run the experiment, and did not recognize that the dry-run it did run had failed.

Generated with Autovoila (Lossfunk) + Claude Code. Human contribution: research proposal, Research Question Sharpener, this critique.